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GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation

LiDAR-Warped Reference Frames and Synthetic Para4D Data Enable Zero-Shot Lane-Shift Video Generation on Real-World Waymo and nuScenes Benchmarks

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Autonomous driving simulators require realistic video from camera poses that were never recorded—a lane shift, a slightly different heading, or a hypothetical trailing vehicle. GeoV2V addresses this challenge by combining LiDAR and depth-based geometric priors with full-video conditioning on the Wan 2.1 video diffusion backbone, generating temporally consistent, geometrically accurate driving videos under large trajectory shifts. Trained entirely on Para4D, a new synthetic CARLA-based dataset of synchronized multi-lane videos, GeoV2V transfers zero-shot to real-world Waymo and nuScenes data, achieving state-of-the-art visual quality and geometric consistency.

AT A GLANCE

Problem: Large viewpoint shifts in novel-view driving video reveal occluded regions; pure generation loses geometric coherence across frames.

Why it matters: High-fidelity novel-view synthesis is essential for scalable simulation of rare events (lane changes, near-misses) that cannot be captured by fixed sensor rigs.

Method: Geometric warping of reference frames via LiDAR/depth priors + Wan 2.1 video diffusion conditioned on warped sequence + Para4D synthetic training data.

Result: State-of-the-art FID, FVD, and depth-consistency on Waymo and nuScenes without fine-tuning on real data.

Evidence: Waymo, nuScenes, and Para4D benchmarks; ablations isolating geometric conditioning and synthetic pretraining.

The Big Picture

The long tail of driving scenarios—rare lane changes, unexpected obstacles, adverse weather—cannot be covered by fixed-route data collection. Novel view synthesis (NVS) addresses this by generating plausible videos from camera poses never recorded. For autonomous driving, NVS must satisfy two requirements that pure appearance-based generation fails to meet: (1) geometric consistency across time, and (2) faithfulness to 3D scene structure rather than statistical plausibility.

GeoV2V grounds a powerful video diffusion prior (Wan 2.1) with explicit 3D geometry, turning the generation problem into a geometry-aware inpainting problem: warp what is visible, inpaint what is occluded, and preserve temporal dynamics.

Why Existing Approaches Fall Short

Image-based NVS methods (NeRF, 3DGS): These methods require dense multi-view captures or long sequences for 3D reconstruction—unavailable in drive-by single-camera settings.

Pure video generation (Sora, Gen-3): Unconstrained video generation ignores 3D structure, producing temporally inconsistent lane layouts and incorrect depth ordering.

Warping-only methods: Geometric warp alone leaves holes where occluded content is revealed by the viewpoint shift; filling them with inpainting lacks temporal coherence across the full video.

Core Mechanism

GeoV2V operates in three stages per generated video:

1. **Geometric warping:** Given the reference video and a target viewpoint trajectory, LiDAR point clouds (or monocular depth where LiDAR is unavailable) project each reference frame to the target pose, producing a warped reference sequence with explicit holes where geometry is occluded.
2. **Conditional video generation:** The Wan 2.1 video diffusion transformer takes the warped reference sequence as a conditioning signal via cross-attention, filling holes and refining geometric artifacts while preserving large-scale 3D structure.
3. **Para4D training:** The model is trained on Para4D, a synthetic multi-lane CARLA dataset providing pairs of synchronized cameras at different lateral offsets under diverse weather, lighting, and traffic conditions—the only source of explicit cross-lane supervision.

Technical Formulation

Let $V_{\text{ref}} = \{I_t^{\text{ref}}\}_{t=1}^{T}$ be the reference video and P_{tgt} the target camera pose sequence. The warp operator $\mathcal{W}$ uses depth D_t and camera intrinsics K to project pixels:

$$\tilde{I}_t^{\text{tgt}} = \mathcal{W}(I_t^{\text{ref}}, D_t, P_t^{\text{ref}} \to P_t^{\text{tgt}}, K)$$

yielding a warped sequence $\tilde{V} = \{\tilde{I}_t^{\text{tgt}}\}$ with per-pixel validity masks M_t marking visible regions.

The video diffusion model G_θ generates the output:

$$V_{\text{out}} = G_\theta\big(z_T, \tilde{V}, M\big)$$

where z_T is Gaussian noise and conditioning on $(\tilde{V}, M)$ is implemented via cross-attention at each transformer layer. The training objective is the standard flow-matching loss $\mathcal{L} = \mathbb{E}[\|G_\theta(z_t, \tilde{V}, M) - V_{\text{gt}}\|^2]$ on Para4D pairs.

Learning or Inference Procedure

Training: GeoV2V is trained entirely on synthetic Para4D data (no real-world data). The model sees diverse CARLA scenes with lateral offsets of 1.0–5.0 m, covering standard lane shifts and extreme viewpoints. Wan 2.1 weights are initialised from a pretrained checkpoint and fine-tuned with the geometric conditioning added.

Inference (zero-shot transfer): On Waymo and nuScenes, only LiDAR-based depth is used for warping; the generator runs without any domain-specific fine-tuning, relying on the synthetic-to-real generalisation of the learned geometric prior.

What the Guarantee Says

The geometric warping step provides an explicit constraint: visible pixels in the reference frame are reproduced at the correct target location with sub-pixel registration accuracy (verified by LiDAR reprojection error). The diffusion model only hallucinates content in provably occluded regions (mask $M_t = 0$), giving a formal separation between geometric reliability and generative uncertainty.

Experimental Findings

- **Waymo lane-shift NVS:** State-of-the-art FID and FVD against prior methods (WoVoGen, GeoFlow, AnyScene), with >15% improvement in FVD at 3.5 m lane shifts.
- **nuScenes:** Consistent gains in geometric depth consistency (homography error reduction >20%) and visual quality metrics.
- **Challenging lighting:** GeoV2V maintains FVD gains in night and adverse weather conditions where geometric priors are noisiest.
- **Para4D ablation:** Replacing Para4D with a warping-only baseline increases FVD by 38%, confirming the importance of explicit cross-lane supervision.

Ablations and Interpretation

LiDAR vs. monocular depth: LiDAR warping improves depth consistency by 12% over monocular depth; monocular depth is sufficient for visual quality (FVD) but not for geometric accuracy (reprojection error).

Conditioning mechanism: Replacing cross-attention conditioning with concatenation reduces FVD by 8%, confirming that the temporal context of the warped sequence is best exploited through attention.

Training data scale: Para4D with 10k scene pairs is sufficient; doubling to 20k improves FVD by only 2%, suggesting diminishing returns from synthetic data.

Reference

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